

Software engineering

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i) SDLC - Software development life cycle

- i) Plan
- ii) Design
- iii) Built
- iv) Test
- v) Deploy
- vi) Maintain

Planning and requirement analysis

- ↳ Define prototype, system requirements
- ↳ Evaluate alternatives to existing systems or prototypes.
- ↳ Research and analyse the needs of end-users

Design:

- ↳ Define user interfaces and interaction flows
- ↳ Specify system interfaces between components
- ↳ Plan network requirement and topology

Build:

- ↳ Product code is build
- ↳ Modules are developed independently, then integrated
- ↳ Coding standard and version control are enforced
- ↳ Unit level checks accompany each completed module

Test

- ↳ Generated code is tested for the given requirements
- ↳ Defects are logged and resolved iteratively

Deployment:

- ↳ Software is certified free of critical bugs
- ↳ Monitoring begins to catch post release issues

Maintenance:

- ↳ Feedback from the market and users is collected
- ↳ Works with feedback to improve the process iteratively.

2) Umbrella Activities- Overview

↳ These activities run in parallel with the entire software process and maintain progress, quality changes, & risk.

↳ Activities are not tied to a single SDLC phase, they span all of them.

↳ Important for keeping large projects under control.

Key tasks of Umbrella activities:

- i) Project tracking and control
- ii) Formal reviews
- iii) Software quality assurance
- iv) Software config. Management
- v) Document preparation

Waterfall model:

- ↳ The Earliest model
- ↳ Also called as Linear sequential (or) Classic life cycle model.
- ↳ Best suited for small projects

Advantages:

- Simple and easy to understand
- Requirement known & easy to manage
- Avoids overlapping
- Well suited for small projects

Disadvantages:

- Poor fit for complex projects
- Not ideal for long-duration projects
- Risks and defects surface late
- High overall project risk if requirements change